

Cardiac Arrest Early Warning Pipeline: Results

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Research prototype. All results are pipeline demonstrations, not clinical validation. Confidence intervals are patient level bootstrap intervals. The synthetic cohort is inflated by design (deterministic drift); the real data test uses 20 Holter patients without photoplethysmogram, blood pressure or vitals, so its role is to bound the electrocardiogram only regime.

1 Synthetic validation

The synthetic cohort has 60 patients with 28 arrest events. Baselines and the temporal model on the patient level validation split:

Table 1: Model performance on the patient level validation split (AUROC).

Model	1 h	6 h	24 h
Logistic regression	1.000	0.954	0.689
Gradient boosting	0.998	0.993	0.911
FEAN temporal model	1.000	0.844	0.667

On the two held out replay patients (one case, one control):

Table 2: Held out replay evaluation (patient level bootstrap CIs).

Horizon	Positives	AUROC [95% CI]	AUPRC	Brier
1 h	12	1.000 [1.000, 1.000]	1.000	0.025
6 h	72	0.891 [0.825, 0.922]	0.785	0.315
24 h	288	0.946 [0.944, 0.948]	0.976	0.137

Table 3: Alert level metrics on the held out replay patients.

Metric	Value
First true alert before arrest	21.7 h
Lead time of true alerts (min, median, max)	0.6, 11.2, 21.7 h
False alarms on the control patient	5 in 24 h (4.9 per patient day)
PPV at sensitivity 0.8 (1 h, 6 h, 24 h)	1.000, 0.307, 1.000
Sensitivity at 1 alert per day (1 h, 6 h, 24 h)	1.000, 1.000, 1.000
Time averaged AUC across lead time bins	0.890

Fusion variants on the same synthetic split (per modality projections plus gated fusion, modality attention or static weights, with explicit presence masks): 6 h validation AUROC is 0.830 for gated fusion, 0.738 for attention and 0.859 for static weights against 0.769 for the

Table 4: Feature block ablations (validation AUROC, 20 epochs each).

Variant	1 h	6 h	24 h	Replay alerts (case, control)
Full	1.000	0.859	0.619	247, 1
Vitals only	1.000	0.755	0.555	38, 0
Vitals plus engineered	1.000	0.845	0.718	51, 0
Vitals plus embeddings	1.000	0.816	0.574	245, 0

concatenation baseline. With the photoplethysmogram removed at evaluation time, gated and weighted fusion stay stable (0.867 and 0.841) while attention fusion degrades to 0.488.

Synthetic patient 2: pre-arrest drift over the last 12 h

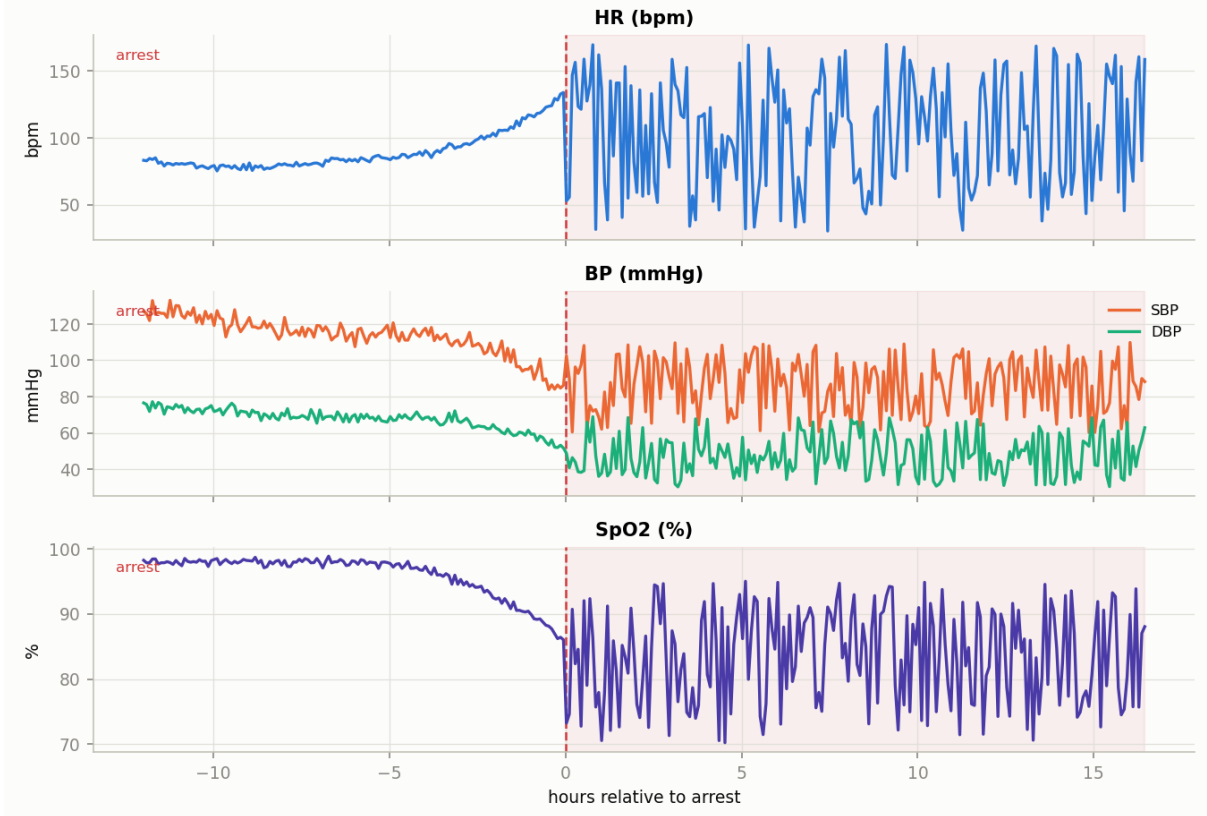


Figure 1: Pre arrest drift of one synthetic case patient over the last 12 hours before arrest (heart rate, blood pressure, saturation). The post arrest region is shaded and excluded downstream.

2 Real data stress test (Sudden Cardiac Death Holter Database)

1829 real five minute windows from 20 records were embedded and evaluated with a 5 fold cross validation grouped by patient. The 24 hour horizon is not evaluable because no recording extends far enough before onset to supply 24 hour negatives. Sequence histories respect record and span boundaries and seeds are fixed; confidence intervals are patient level bootstrap intervals.

The negative result is genuine and not a training failure: train loss decreases across epochs while held out loss stays flat, and predictions spread widely, yet held out records remain at or below chance at 6 hours for every variant. On one fold several variants become confidently inverted (AUROC near zero), learning record specific patterns that do not transfer across 16 training patients.

Synthetic patient 2, window at $t = 38.2$ h

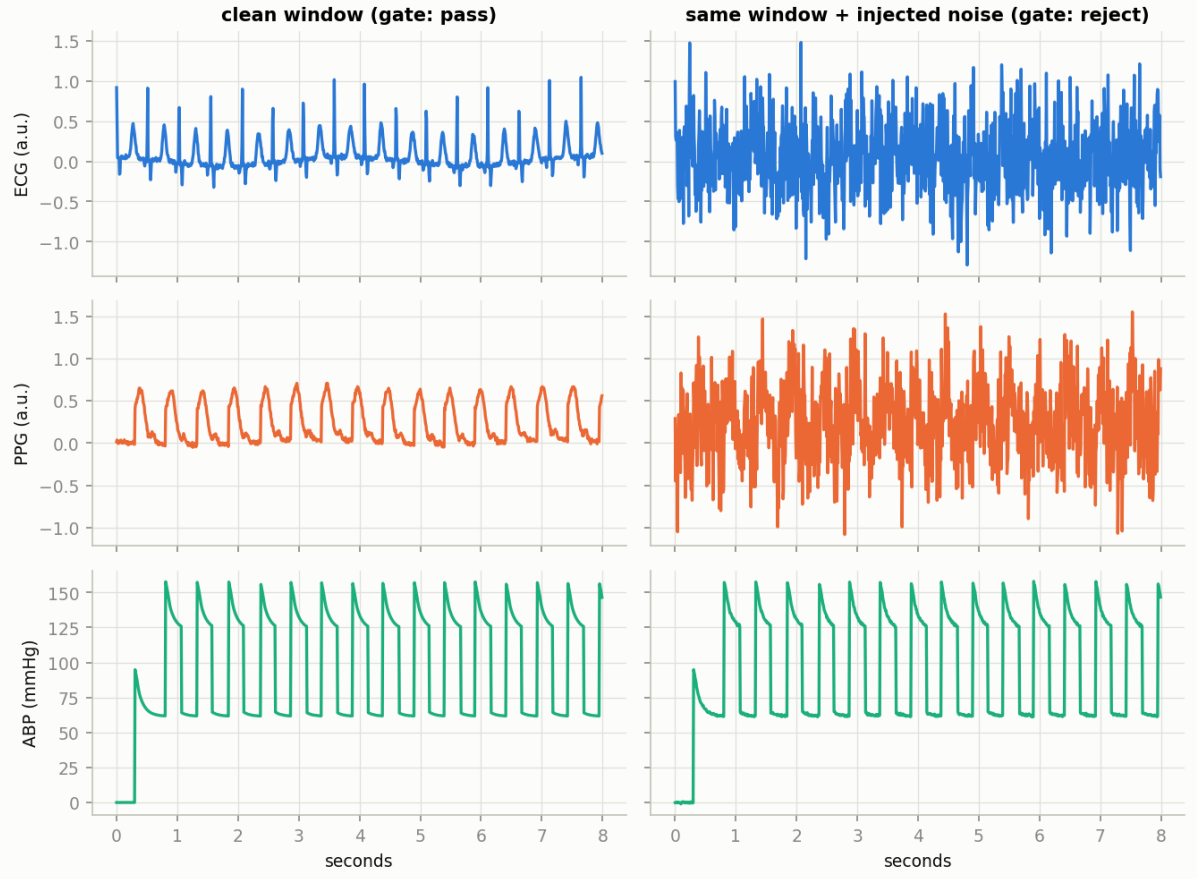


Figure 2: One 30 second window: clean electrocardiogram, photoplethysmogram and arterial pressure (left) versus the same window with an injected noise artifact (right), which the quality gate rejects.

Variant level lessons. The models do recognize patients: a linear identity probe on the context vector classifies held out train windows into patients at 0.13 to 0.47 accuracy across variants (chance 0.05). The identity signal is carried by the embedding block (probe 0.439 without engineered features versus 0.130 without embeddings). Identity adversarial training helps slightly at both horizons but does not reduce the probe (0.423 to 0.470), so gradient reversal does not remove the leakage; modality attention is the only architecture that substantially reduces it (probe 0.143), and it is also the best held out variant at 6 hours. Concatenation, gating and static weights are statistically indistinguishable; modality attention is the best deep variant at 6 hours but also the least stable. Missing modality masks keep predictions finite when streams disappear but do not change the ranking. The 1 hour signal is carried by engineered heart rate variability and ectopy features (0.767 without embeddings), and the ECG FM embeddings dilute it (0.632 full model). Attention salience shows the model spreads its attention over history and puts near zero weight on the current window.

3 Slow risk layer

QT interval, T wave alternans index, QRS width and long term heart rate variability markers are aggregated over 3 hour blocks and scored with a logistic regression (59 blocks, 16 records, grouped cross validation, 0.5 hour label margin rule). The layer stays below chance on this data: 1 hour AUROC 0.426 [0.193, 0.667] with 5 positives, 6 hour AUROC 0.313 [0.190, 0.400] with

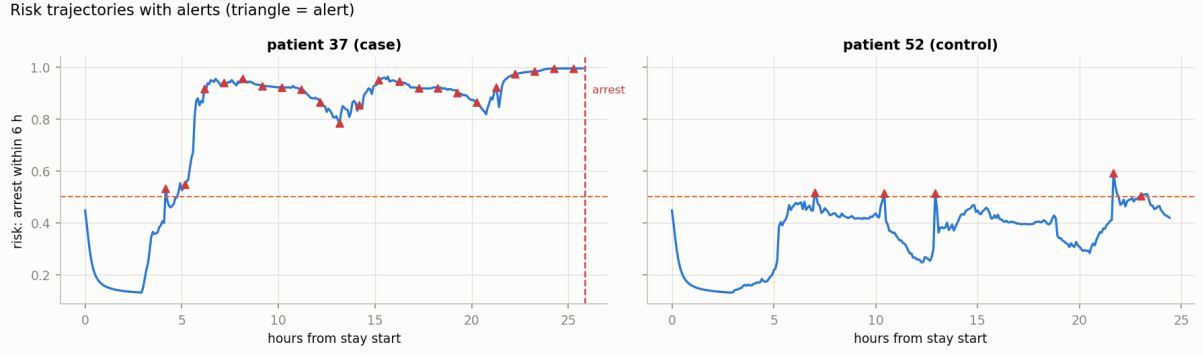


Figure 3: Risk trajectories of the trained pipeline on the two held out patients. Triangles mark alerts, the dashed line is the threshold. The case patient (left) first crosses the threshold about 21 hours before arrest; the control (right) produces a small number of false alarms.

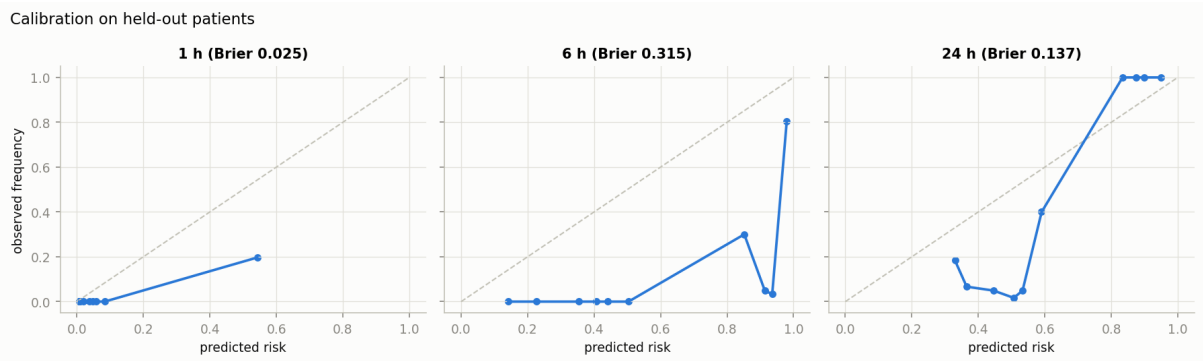


Figure 4: Calibration curves per horizon on the held out patients. Predicted risk versus observed frequency; the diagonal is perfect calibration.

28 positives. Slow electrocardiogram markers alone do not separate pre arrest blocks from far blocks on Holter recordings, consistent with the short horizon result.

4 Robustness

Five perturbation families were applied to the held out replay patients and scored with the three trained models (concat versus mask aware gate and weight). Waveform perturbations were re embedded through the real encoders: ECG noise at SNR 20, 10 and 5 dB, baseline wander, 50 Hz mains, flatline halves, saturation, and PPG noise at 5 dB. Feature level stress covered vitals noise at 0.5 to 2 times the per column standard deviation, missing PPG, delayed measurements shifting the whole vector by 5, 10 and 15 minutes, and a sensor failure where PPG dies mid stay.

Every perturbation lands within 0.03 of clean. The only directional effect is the delay family, which degrades monotonically but mildly (concat 0.929 to 0.920 at 15 minutes). Missing PPG, sensor failure, vitals noise and all ECG artifacts leave the score unchanged, and the mask aware models behave like concat here because synthetic PPG carries little signal.

Honest caveats: this measures robustness on synthetic data, where the risk signal lives in the vitals and engineered blocks (the earlier ablation finding), so waveform robustness partly reflects the encoders and partly that the embeddings matter little there; the quality gates would reject flatline and saturation windows in the real pipeline, and these tests bypassed the gates and still held; and with 2 replay patients these are point estimates.

Table 5: Real data stress test (20 patients, 1829 windows).

Model	1 h AUROC	6 h AUROC
Zero shot (synthetic trained model)	0.338	0.447
FEAN, concatenation (baseline)	0.632 [0.516, 0.748]	0.402 [0.344, 0.456]
FEAN, identity adversarial	0.697 [0.599, 0.792]	0.436 [0.368, 0.501]
FEAN, identity adversarial + PCGrad	0.663 [0.557, 0.781]	0.459 [0.382, 0.537]
FEAN, gated fusion + masks	0.655 [0.546, 0.778]	0.386 [0.337, 0.435]
FEAN, modality attention + masks	0.638 [0.519, 0.754]	0.473 [0.402, 0.538]
FEAN, static modality weights + masks	0.660 [0.546, 0.776]	0.394 [0.347, 0.443]
Ablation: engineered features only	0.767 [0.705, 0.837]	0.508 [0.461, 0.555]
Ablation: embeddings only	0.707 [0.602, 0.812]	0.422 [0.377, 0.462]
Slow risk layer (3 h blocks, QT/TWA/HRV, LR)	0.426 [0.193, 0.667]	0.313 [0.190, 0.400]

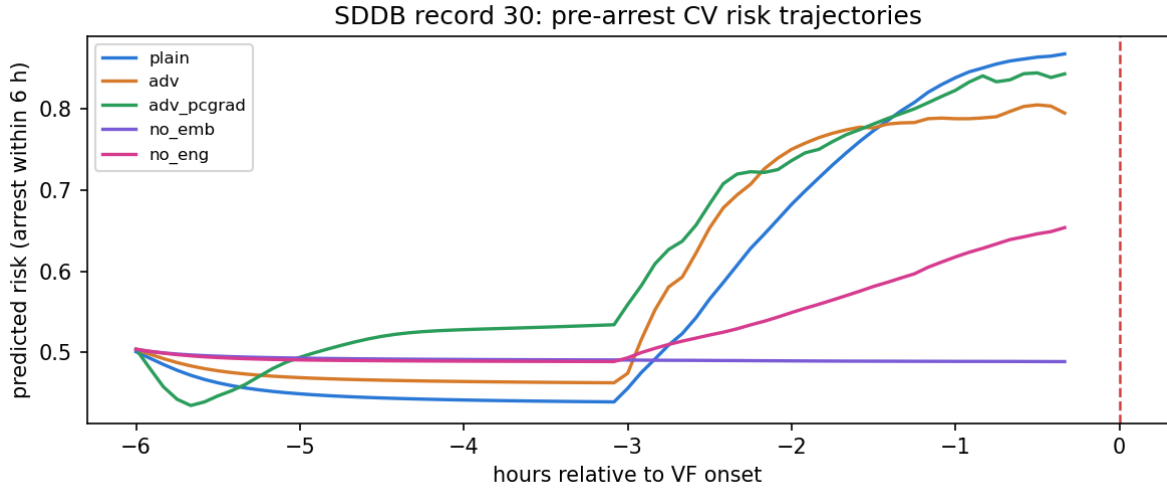


Figure 5: Cross validated risk trajectories for one real patient, hours relative to ventricular fibrillation onset. The predicted risk does not rise before the real event in any variant.

5 Summary of findings

The pipeline machinery is validated end to end on synthetic data. The synthetic trained model carries nothing to real electrocardiograms (zero shot at chance). No model variant trained on 20 real patients beats chance at 6 hours. At 1 hour a weak signal exists and comes from engineered rate features, not from the foundation model embeddings. The decisive next step is the credentialed MIMIC III evaluation with full modalities.

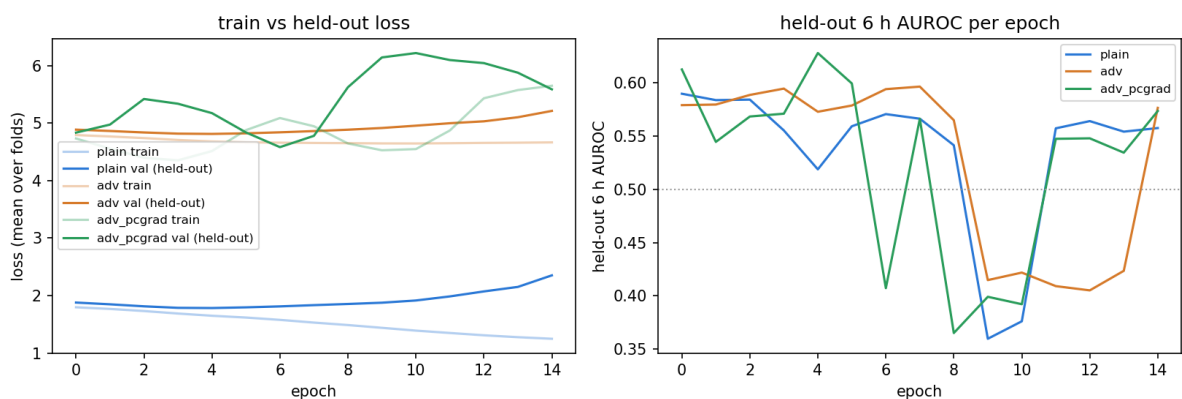


Figure 6: Train versus held out loss per epoch (mean over folds) for the three identity adversarial variants, and held out 6 hour AUROC per epoch.

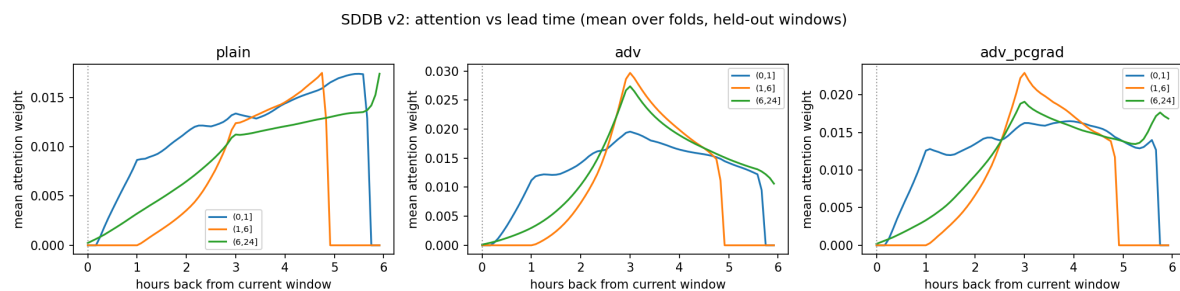


Figure 7: Attention salience by lead time bin (mean over folds): the model spreads attention over history and puts near zero weight on the current window.

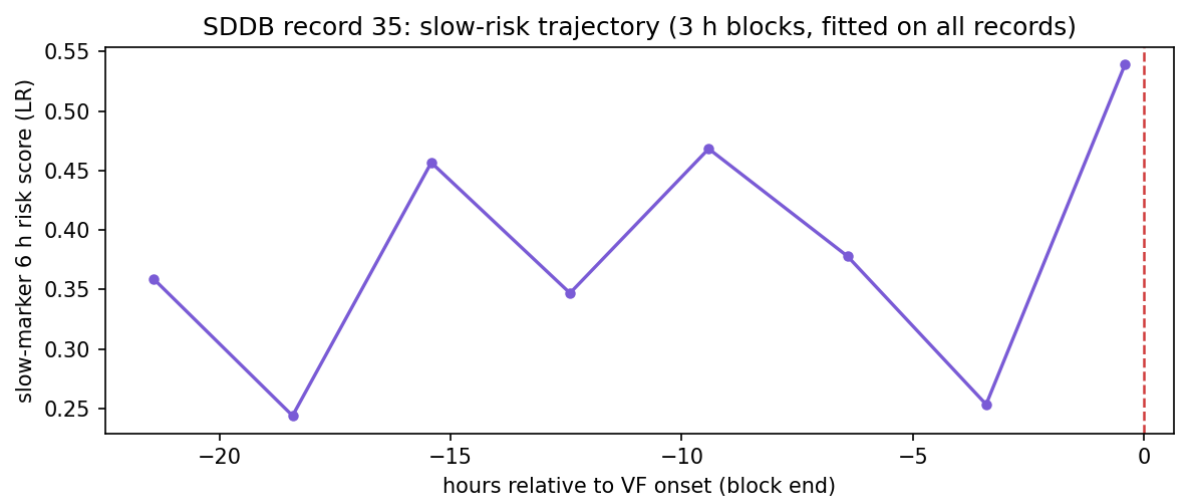


Figure 8: Slow risk score per 3 hour block for the record with the longest pre arrest region (logistic regression fitted on all records, illustrative). The score does not rise as the event approaches.

Table 6: 6 h AUROC under perturbation (clean baseline 0.929 to 0.945).

Perturbation	concat	gate	weight
Missing PPG	0.937	0.952	0.947
Delay 5 min	0.926	0.943	0.941
Delay 10 min	0.923	0.942	0.940
Delay 15 min	0.920	0.941	0.938
Sensor failure (PPG dies mid stay)	0.929	0.945	0.943
Vitals noise 2x std	0.929	0.951	0.943
ECG noise 5 dB	0.934	0.949	0.945
ECG flatline halves	0.928	0.940	0.952
ECG saturation	0.929	0.946	0.943
PPG noise 5 dB	0.958	0.949	0.963

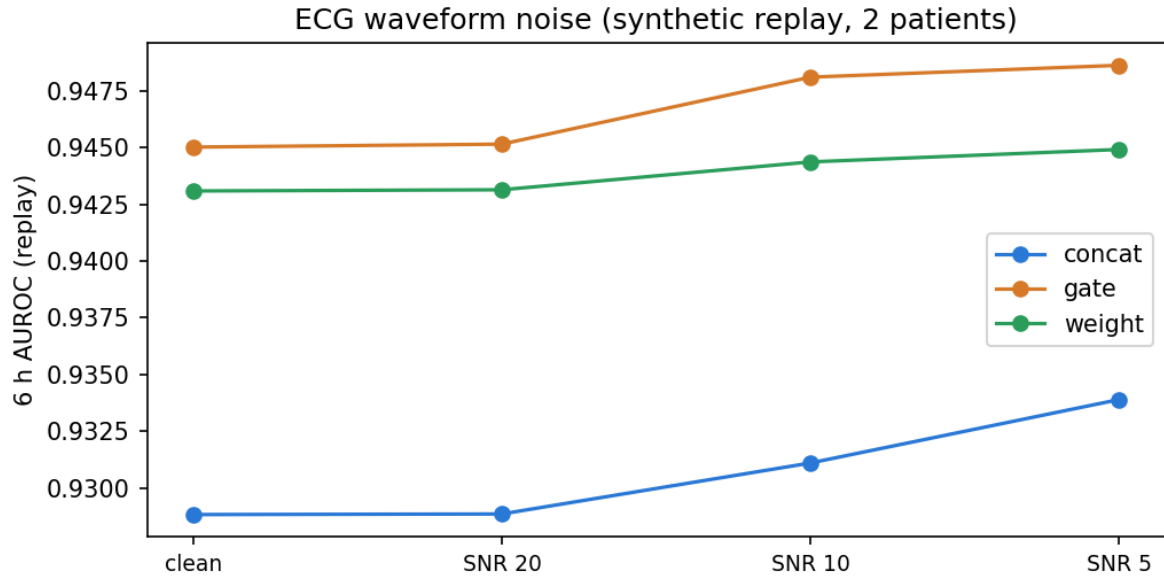


Figure 9: 6 h AUROC versus ECG waveform noise level (synthetic replay). No degradation down to SNR 5 dB.

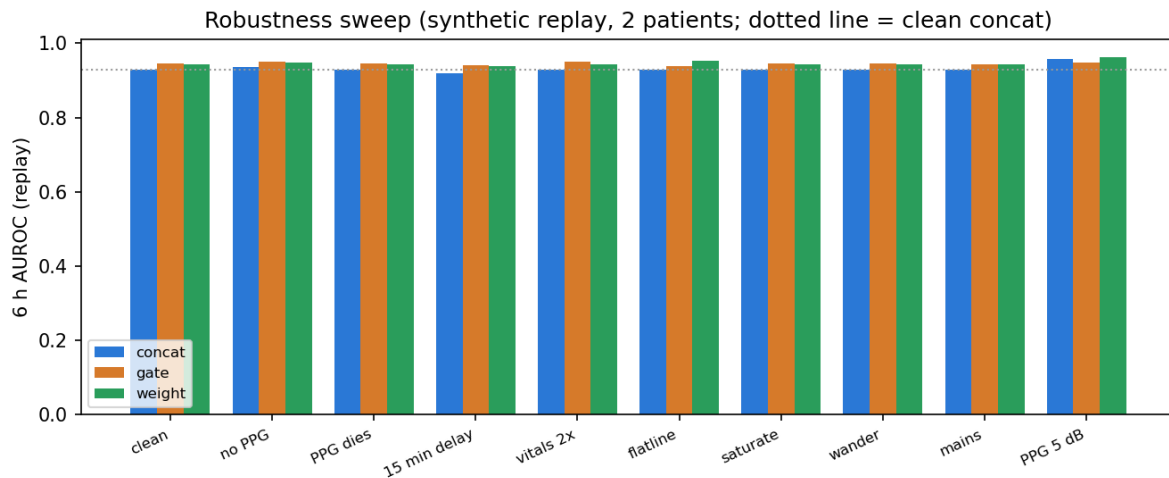


Figure 10: Robustness sweep across all perturbation families (synthetic replay; the dotted line is the clean concat score).